



Radar for Autonomous Driving – Paradigm Shift from Mere Detection to Semantic Environment Understanding

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1 Introduction

The challenges for sensors and their correlated perception algorithms for driverless vehicles are tremendous. They have to provide more comprehensively than ever before a model of the complete static and dynamic surroundings of the ego-vehicle to understand the correlation of both with reference to the ego-vehicle's movement. For dynamic objects, this means that radar has to provide the dimension and complete motion state as well as the class information, in highway, rural, and inner city scenarios. For the static world, new algorithm schemes have to be developed to enhance the shape representation of an object by image like semantics. In order to generate the necessary information, radar networking for 360° coverage have to be reinvented. Radar data processing toolchains have to be revolutionized by applying artificial intelligence and advanced signal processing in a synergetic manner.

The paper will provide an overview of state-of-the-art automotive radar usage, deduces future requirements for highly automated driving and will present recent advances in radar based environmental perception.

1.1 The History of Automotive Radar Answers the Question: “Why Radar?”

Automotive radar has already reached a market penetration that leads to several tens of million units used. It has grown up to a status where it has found its way into nearly all car manufacturers portfolio in the world. They are used in all vehicle types: from premium vehicles to even the smallest sedan platforms and from vans up to heavy trucks and travel busses. With the introduction of the collision prevention assist®, radar sensors have become even standard equipment in passenger cars [1]. The major reason for the success story of automotive radar is its physical principle that offers unique performance features at reasonable costs. Among others, they are independent from environmental factors like illumination and weather conditions, are capable of

directly measuring positions, material parameters and Doppler velocities, have multiple field of views and are easily compatible to vehicle designs and paintings. The packaging benefit allows a 360° coverage from the vicinity to extreme far ranges. Radar performs under conditions where other sensor types fail or suffer in performance and it can virtually look through vehicles by exploiting reflections between the road surface and the vehicle's underbody and hence makes the invisible visible. Very often, the radar signal is the only remaining signal out of the total perception system the controller sees in adverse environment conditions or in the dark.

Over decades, the performance requirements increased steadily from simple detection and ranging tasks in blind spot monitoring or cruise control systems up to smart environment perception tasks for present day semi-autonomous evasion and braking functions [2]. However, the utmost push in performance requirement is initiated with the trend towards highly automated driving and fully autonomous driving [3–6]. In 2013, the first stride ahead towards higher automation has been made with the fully autonomous Bertha drive of a Mercedes-Benz research sedan [3–6]. The major lesson learnt was that future automotive radar systems have to provide imaging like capabilities, i.e. very high resolution in space and velocity, and have to interact in radar networks composed of four or more short-, mid-, and far range radars, which allow for 360° highly comprehensive perception tasks. This insight is backed by the published sensor set-ups of other groups developing driverless vehicles [7–9]: a from year to year steadily increased number of radar sensors in their setups can be noticed.

1.2 Paradigm-Shift in Development Guidelines: Deep Learning and Model-Based Approaches Together on Imaging Like Radar Data

In general, sensors and perception algorithms for driver assistance and active safety systems are designed to observe the direct neighbours of the ego vehicle and to identify the relevant objects. They are designed for specific tasks, where human drivers often struggle: these are either situations where constant awareness and quick reactions are necessary or where the driver has to estimate distances and velocities accurately. Example applications are lane keeping, emergency braking and parking assistance systems. However, human drivers excel any of today's assistance systems in tasks, where contextual understanding of the surroundings is needed, e.g. the anticipation of other road user's driving behaviour, reaction to lost cargo or path planning in construction sites. Driverless cars have to cover all activity areas and ideally under all conditions. Thus, sensors and perception algorithms for driverless driving vehicles face a much harder task. They have to describe the complete static and dynamic neighbourhood of the ego-vehicle and have to understand the interplay of both. Considering the natural benefits of radar lined out above, it is logical that on a global scale radar becomes the back-bone of driverless vehicles. Therefore, radar sensors and their perception algorithms have to be further developed from a pure detection and ranging system, to a complete semantic environment understanding system [10–12]. One important consequence is that radar signal processing has to be complemented with machine learning and scene understanding concepts to keep radar in the leading edge of remote sensing. A pre-requisite for applying new algorithm concepts is that

radar data becomes imaging like, in order to provide many valuable detections per object. Hence, future radars have to provide high resolution and accuracy in space and velocity [10, 12, 13]. Radar systems developed for level 1 to 3 driving tasks will fail in providing the required information [14].

1.3 Is Deep Learning Really the Answer to Everything?

The intensive and wide spread use of deep learning methods for classification and other tasks in the automotive area has been initiated by their successes in image recognition tasks. It was pushed by various companies like Nvidia, who provide new automotive chips and powerful GPUs, Google, Microsoft and Apple, who created deep learning software packages and sophisticated algorithms. Until now, radar took advantage by exploiting its unique characteristics, gaining the required information by model based approaches using well-known physical relationships. Although a lot has been achieved using those techniques, the success that has been reported using deep learning requires radar developers to adapt to new methods in order to push their achievements even further.

End to end approaches exist which ignore all signal processing steps, but they require huge amounts of labelled training data and computing power. The main disadvantage of deep learning algorithms is that it is very hard to determine their behaviour in unknown situations, which are not present in the training or test dataset. Therefore, if a model-based solution to a problem is well known, this model is for us almost always preferable to a deep learning method. For example, it seems unlikely that well established signal processing methods like Fourier transforms or peak detection algorithms will be replaced by Deep Learning, because a clear task and a working method are already at hand.

Then again, on some real world problems it is virtually impossible to determine a good model. For example, humans can easily classify objects as a car, but it is difficult to describe the classification task in a model. Variety in colour, the number of visible wheels, occlusion or even a cover with a tarpaulin exemplify the difficulty of the task.

For these tasks, deep learning can help to achieve the required performance, because the generated model is so complex that it cannot be designed by a human. It seems therefore logical that traditional and deep learning techniques should cooperate, to enhance their specific strengths.

Unfortunately, an easy adoption of all available techniques and tooling ecosystem fails due to the totally different nature of radar data compared to vision systems. Hence, research activities concentrate among other things on getting the right data representation to benefit from the available ecosystem and to apply radar data to deep learning techniques.

The first group that applied deep learning to radar to solve the problems of semantic classification of static and dynamic objects as well as the estimation of the course of the road using radar sensors, was Daimler research [15–20].

The class information of dynamic objects can be used to choose an object-specific tracking method. This is necessary since the motion pattern of e.g. pedestrians, cyclists, cars or trucks differs significantly and therefore custom motion models are

essential. If mirror effects, ghost targets and other sensor artefacts are annotated correctly as such and algorithms further down the processing chain can rely on semantic class labels for each measured reflection, all these methods become much easier and for example tracking algorithms can perform better and faster.

The fusion of static and dynamic object classification generates a thorough understanding of the traffic scenario and therefore critical situations can be anticipated at an early stage and vulnerable road users can be considered with particular caution. In addition to classification, deep learning makes also road course detection possible, so that for example even on snowy roads path planning with radar data is possible.

Classification of static objects improves landmark detection and validation which in turn optimizes the self-localization and allows driverless parking even in parking garages [21, 22]. For example, movable objects like parked cars or trash cans are rejected as possible landmarks so that only truly static objects are used for localization. Semantic information of static objects can also be utilized for associating the radar landmarks with landmarks from third party maps and thereby localizing the ego vehicle on nonradar specific maps.

The subsequent three sections will provide examples which underline the potential of artificial intelligence and deep learning in model based concepts for radar perception. The Sect. 2 focuses on radar applications for dynamic objects and give detailed information on how moving objects can be semantically classified and how this information can enhance tracking algorithms. In the Sect. 3, we turn to the static world: here we introduce semantic radar grids and their application in classification and road course estimation. Finally, we give an outlook to high resolution algorithms which motivate the development of high resolution radar to enhance the aforementioned applications.

2 Radar Application for Dynamic Objects

2.1 Classification

The eponymous applications for radar – detection and ranging – allows to make basic statements like: “In distance x is an object”. If additionally Doppler velocity, azimuth angle and reflected energy can be obtained from the sensor, the claim can be formulated: “In distance x and direction ϕ is an object with radial velocity v and that object reflected the energy amount E ”. However, the nature of the detected object, that is the *semantic class* the object belongs to, remains unknown. The aim of classification algorithms is to specify the “what” of the detection: Does a measured reflection belong to a moving pedestrian, a bicyclist, a truck or a car?

Semantic knowledge about the detection can be utilized in many scenarios: class specific tracking algorithms can easily outperform general tracking approaches, path planning for autonomous vehicles has to take information about the surroundings into account, and critical situations may be reduced or even avoided if other road users are identified early on.

Only recently, this new application of radar sensors became more popular and gained attention especially in the automotive field [18, 23]. This timing is not a coincidence

but tightly coupled to the rapid improvements in the fields of machine learning and image understanding. The underlying methods and algorithms – which were originally designed for image-like data – can also be translated to the radar world. This translation step, i.e., the modification of the algorithms which work seamlessly on camera images to radar data, is by no means trivial or straight-forward and is still ongoing research.

The difficulties already start at the labelling stage: Almost all machine learning approaches rely on manually annotated data, so called “labelled data”. During a training stage, neural networks or other machine learning algorithms learn from this annotated data and are then able to apply this learned knowledge to new unseen data. For images, large publicly available databases with millions of annotated samples exist and it is therefore easy to start with the optimization or development of new algorithms. For radar data, no such free dataset exists. The reasons for the lack of publicly available annotated radar data are manifold. Firstly, radar data with ego-motion compensated Doppler velocities is harder to obtain than camera images, which everyone can capture with their smartphones. Secondly, one has to decide on which data level the labelling should be performed: after receiving the reflected wave, various pre-processing steps like fast Fourier transforms and peak detection algorithms are applied and for each intermediate representation of the data, arguments can be found for and against labelling at this stage. Thirdly, the way radar sensors perceive the world is very different from the way humans do. Therefore, human annotators have to be trained to work on radar data so that they understand some of the sensor properties. Experience showed that humans can easily identify traffic signs in a camera image but struggle to identify radar reflections at given distances and azimuth angles that belong to the same object. In addition to these challenges, misalignments of the sensors, time synchronization problems with the ego-motion, multipath reflections, mirror effects and ghost measurements increase the difficulty of the labelling task.

To achieve our aim to classify measured radar reflections, we developed our own labelling software, collected data, defined a class hierarchy and labelling instructions, and trained labelling experts who annotate the data. Our current dataset for classification of dynamic objects contains measurements from over 4.5 h of driving, that is, over 100 million radar reflections were collected of which 3 million belong to 6200 different moving objects. For each of the 100 million reflections an annotator had to decide if said reflection belongs to a dynamic object and if so, to which class this object belongs to.

To use the labelled data for training of our deep learning algorithms, some pre-processing is necessary. In a first step, the clustering, all radar reflections that belong to the same object are grouped together. The formed clusters are then the basis for the feature extraction. In this second step, statistic properties and other descriptive quantities are calculated over all reflections in a cluster. The selection of meaningful features is a non-trivial task since it can be easily proven that the approach “the more features the better” is actually harmful for the classification scores. At the same time, one has to keep in mind that manually created features have only a certain ability to generalize to unseen data.

Our current approach is depicted in Fig. 1. We use a special kind of neural network, a so called long short-term memory network (LSTM), which is able to learn from a sequence of input data and can hence take the temporal evolution of an object into account. These kind of networks are also used by popular devices such as Amazon’s Alexa.

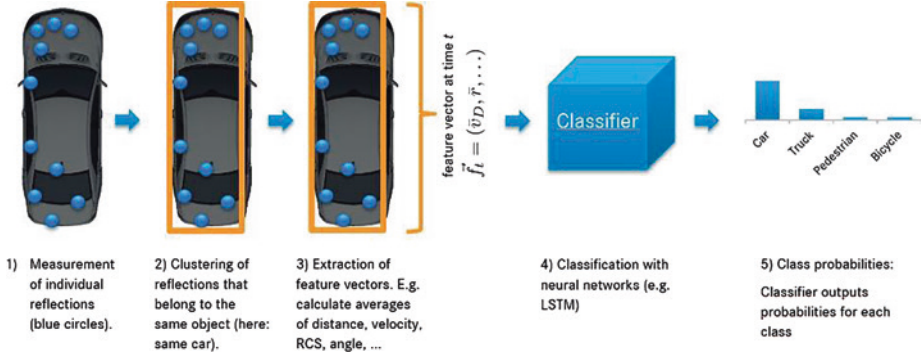


Fig. 1. Classification pipeline. From left to right: measurement of targets, clustering, feature extraction and application of classification algorithms.

One may wonder why machine learning methods are necessary at all for the classification tasks and whether “classic” algorithms which do not rely on millions of labelled reflections would not work just as good. In Fig. 2, different examples of reflections that belong to individual cars are displayed. One can easily notice that shape, extension and distribution of the reflections differ greatly among the examples. Additionally, one can find clusters of pedestrian groups whose distribution of reflections looks similar to those of a single car. Simple algorithms that only target the shape of a cluster and the measured Doppler velocities cannot generalize well enough to deliver reasonable classification results without creating many false positives. The complicated decision rules that are generated by a deep neural network cannot be easily (re-)produced by “classic” algorithms and history has shown that deep learning excelled all other approaches in this kind of tasks.

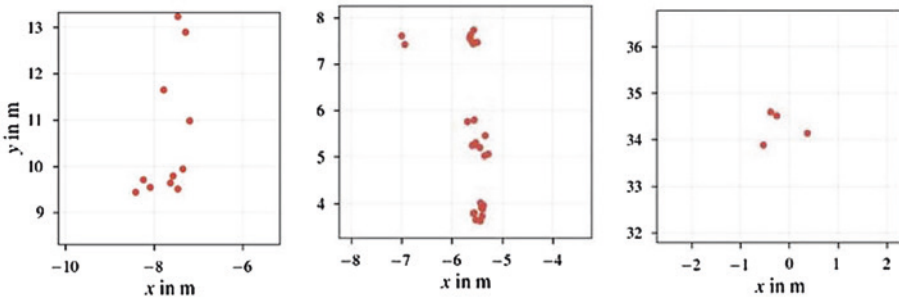


Fig. 2. Three real world examples of radar reflections from three different cars. Data from one sensor, accumulated over two cycles, is shown. With greater radial distance, the number of measured reflections decreases and hence the classification accuracy also declines.

The last ten years in the field of scene understanding with camera images displayed impressively that a classifier that generates its own features outperforms any algorithm that can only work on a pre-defined feature set. It seems therefore desirable to build classifiers that take the raw radar data as input and create their own descriptive features with which the classification can be done. This step is still ongoing research but first experiments at Daimler research showed promising results so that semantic understanding of moving objects using only radar is in reachable distance.

2.2 Extended Target Tracking

Target tracking algorithms play a vital role in the fusion and behaviour planning activities in the autonomous driving functions. Earlier advanced driver assistance systems and some limited autonomous functions [5] have managed to carry out their safety critical functions by performing point based target tracking. The target extensions such as length and width are approximated heuristically by the algorithm since automotive sensors of those days did not had sufficient resolution. As the resolution of the sensors increased in parallel to the affordability in terms of purchase cost, this factor enables us to improve the target tracking applications w.r.t estimation of target size and shape, yaw-rate and orientation. This allows us to program and model higher autonomy driving functions. Implementing a real-time automotive extended target tracking algorithm [24] is very closely tied with the availability of computing resources, communication band-width and allowed operational latency in the processing chain.

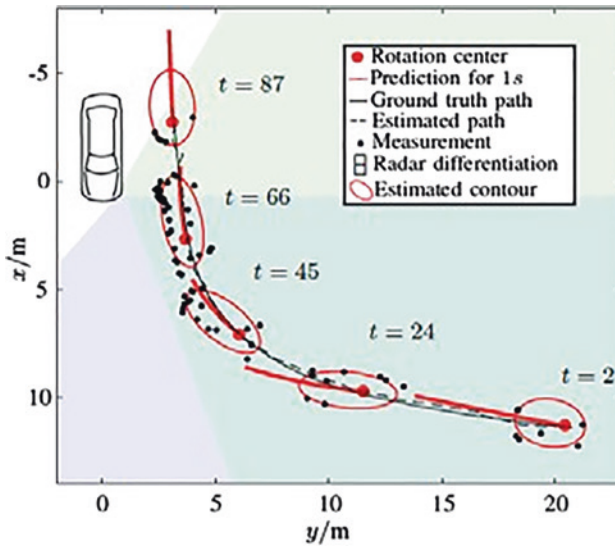


Fig. 3. As described in [25]. The estimated shape is approximated using an Ellipse.

It is vital to understand the important use-cases that necessitate a robust and stable sensor-set specific extended target tracking algorithm: all overtaking manoeuvres, collision avoidance and tight corner manoeuvres, tracking of group of pedestrians, bicyclists, targets of unidentified type and form.

Before we go ahead with discussion of estimating the size of the target under observation, it is noteworthy to consider parameters such as the velocities in x and y direction, as well as the yaw rate: $(\dot{x}, \dot{y}, \dot{\psi})$. These parameters are estimated [25] by building knowledge about the Doppler velocity distribution of the different observation across the target vehicle. The higher the radar resolution is in space and velocity, the better and more robust the Doppler profile can be deduced and the yaw rate calculated. Prior shape models play an important role in robust estimation of shape of the target. As shown in Fig. 4 and as derived in [26], a prior spatial model along with Doppler profile called Volcanormal model was proposed. This model uses a symmetric positive definite 2-d Ellipse to describe the size of the target under observation. The estimated Doppler profile is used to determine the yaw rate $\dot{\psi}$ which enables us to predict the orientation of the target under observation.

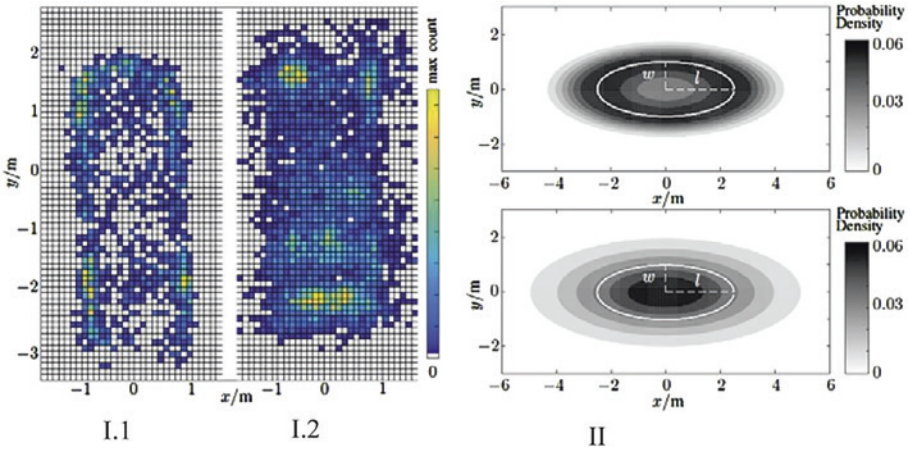


Fig. 4. As described in [25]. Sub-fig (I.1) shows an aggregated radar measurement grid with 0.1 m cell resolution for a right-turn scenario (duration: 7.6 s). Sub-fig (I.2) displays a grid of a different urban driving scenario (duration: 47.0 s) and sub-fig (II-top) shows our radar spatial distribution, called Volcanormal, with reference to a usual normal distribution (II-bottom).

Once the prior model and Doppler profile is derived using either one or more frames of measurements as shown in Fig. 3, the final step of measurement differentiation is to calculate the measurement delta's, a Doppler-fit and the shape or the contour of the target under observation is estimated based on the measurement spread and $\dot{\psi}$. The parameters length and width can be extracted using the Ellipsoid contour information of the observed target. The effectiveness of this approach is displayed in more detail in [26].

2.3 Classification Assisted Tracking

Classification assisted tracking (CAT) uses the information obtained from the classification algorithms as an important source in the information gain likelihood calculation in addition to the kinematic information of the measurement and state space. This enables us to perform robust tracking in a dense target and clutter zone. Target separation ability of the tracking algorithm is enhanced to a very high success rate. This subject is discussed in detail in [27] and [28]. CAT uses traditional Bayesian probabilistic framework with different hypothesis and combinatorics nodes to derive its different likelihoods. However, a numerical approximation suitable for a real-time implementation is highly possible.

3 Scene Understanding of Non-moving Targets

Only until recently, the static world, i.e. reflections from non-moving targets, were solely considered as obstacles – if taken into account at all

With the demand for more advanced driver assistance systems, a better perception of the world is needed and hence also semantic information about non-moving objects is necessary.

Although radar data is sparse in comparison to LIDAR or camera, a lot of information can be inferred if it is accumulated over multiple times steps and radar physics is exploited. To this end, occupancy grid maps specifically designed for radar data can be used.

In Fig. 5 an occupancy grid can be seen, which depicts the static objects around the ego-vehicle. Probability of occupancy of individual cells is colour coded with yellow depicting a high probability and blue the state of free space.



Fig. 5. Occupancy grid map centered around the ego-vehicle.

3.1 Semantic Radar Grids

Semantic knowledge enables us humans to act appropriately. We can find a proper parking spot, even if no road markings are present. We know that we have to drive carefully in areas where parked cars block the view to the side walk where children are playing. We identify our position on proper landmarks, like buildings, trees and the road course and do not take parked cars and other movable objects into account.

For dynamic objects, the distribution of the Doppler velocity is a characteristic radar feature, to distinguish between different object classes. This cannot be used for the static world. Therefore, the distribution of the positions of measurements and the intensity of the backscattered signal are key feature elements, which are accumulated in the radar grids [15–17, 20].

Deep learning methods from image processing, as used for example in the “Large Scale Visual Recognition Challenge”, can be adopted to extract semantic information in radar grids. Deep convolutional neural networks (CNNs) have been proven capable to learn models for a variety of different data from many fields. Hence, using the right preprocessing steps, this technique can also be used for radar. One appealing property of this approach is that not only information of one object is taken into account but also the object’s surroundings are considered.

Even though objects have a very different appearance in radar than in images and features we are familiar with are missing, e.g. color, a lot of advantageous properties can be used to get the right classification result. Because distances are measured directly, the radar sensor does not suffer from perspective distortion, so that the real size of an object can be determined. Additional material properties can be detected by measuring the radar cross-section. This very different perception of the world is unique to radar and hence situations can be resolved in which cameras are fooled by mirrors, monitors or printed images.

For the training of deep learning algorithms, an annotated dataset is needed. Just as in the case of radar data for dynamic objects, no public dataset is available and therefore we created our own dataset.

To build up the dataset efficiently, the following labeling process is pursued: data is collected in a grid whose cell positions are given in a world coordinate system and hence reflections from static objects, measured at different times, fall into the same area in the grid, irrespective of the motion of the ego-vehicle. It is then possible to assign class labels to regions in the grid and thereby labeling all reflections of the same object at once. With this approach, over 50.000 real world objects were labeled, yielding several millions of training examples from different measurement times.

We proved that classification of different object classes like cars curbstones, trees, poles and plants is possible. Because grids – other than most images – have no natural orientation, all rotations of a grid have to be learned [15–17, 20]. An example of a semantic grid map in which each pixel is assigned to a class is shown in Fig. 6.



Fig. 6. Semantic radar grid map. Same scene as in previous figure. Colors indicate class affiliation (Color figure online)

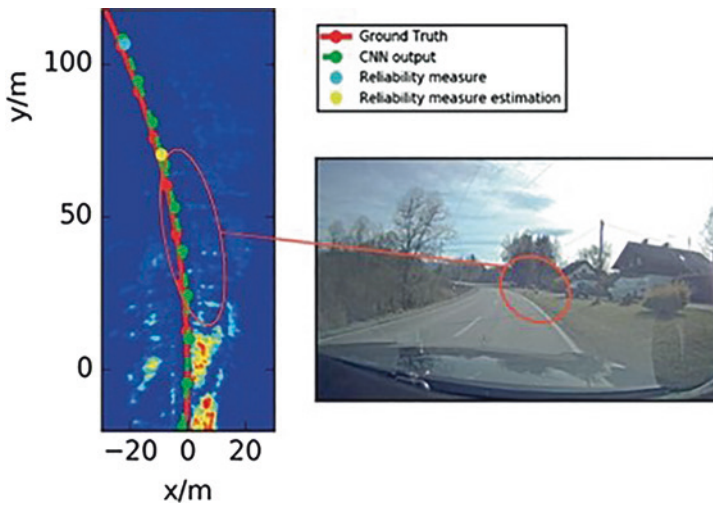


Fig. 7. Example of road course estimation. Front facing documentation camera on the right, occupancy grid with label, road course estimation, reliability measure and its estimation on the left. (Color figure online)

3.2 Road Course Estimation

One of the essential tasks for an autonomous vehicle is to gather knowledge about the road path ahead. With this information, it is possible to plan the path of the future trajectory and to verify a digital map. Radar sensors are especially useful for this purpose because of their ability to detect targets even in high distances which is needed to capture the geometry of the road.

Radar sensors mounted on a vehicle are unable to receive reflections from the road or lane markings directly, and hence the course of the road has to be found indirectly through the surrounding areas. Objects like guard rails, reflector posts or even the transition from tarmac to the road's surroundings (e.g. grass on the side of country roads) can be used to draw conclusions on the most likely road course. Since the reflections from these objects are relatively sparse, especially from the edge of the road, it is beneficial for further processing to transform the received radar data into occupancy grids to accumulate the radar targets and filter them in the time domain.

In the literature, the road course is most commonly estimated by applying rule based algorithms on occupancy grids [29]. These algorithms are built on a few assumptions and mostly on model based road courses. The main assumption is that the road border objects are parallel to the road course while the most commonly used road model is that of a clothoid or an approximation of a clothoid through a cubic polynomial in lateral direction. Yet these assumptions don't always hold true. The road border is not always parallel to the course of the road or the lane the car wants to stay in. This assumption is for example violated at junctions and highway exits. Also the assumed model of the road course isn't always accurate. At the beginning of a curve on a rural road, there is a transition from one clothoid to another which cannot be approximated accurately by a single one. It would be necessary to detect these cases where the assumptions are violated and then be handled separately.

Since this approach is not very practical, we looked for a way to solve the problem with a single procedure. As described in the last section, radar occupancy grid maps allow us to transform the measured reflections into a shape which modern convolutional networks can take as an input [20]. Training data for road course estimation can be easily generated with a high precision localization system. With such a system, the position of the lane the car is driving in is known in the occupancy grid and thus, labelled data is created automatically while driving.

For the use in an automated application, not only the road course has to be estimated, but it also has to be known, how reliable this estimate is. The use of CNNs shows another advantage in this regard, since such a reliability measure can be trained simultaneously with the regression. To train the neural network, we recorded over 700 km of rural roads and highways in southern Germany. Figure 7 shows an example of the test set of a rural road with a junction. Classical algorithms will fail in such a case because of the large scattering area on the right hand side (see red markings in the figure) of the road which leads to a not recognizable road border. Neural networks are able to detect the specific conditions in a scene and react accordingly.

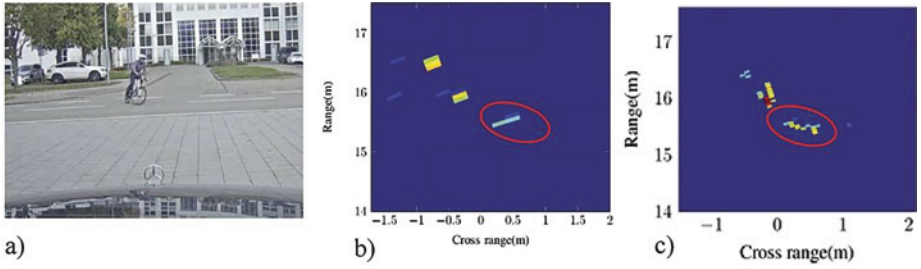


Fig. 8. (a) Camera view of traffic scene. (b) Measured reflections without high resolution. Only few reflections of the bicycle are visible. (c) Highly resolved radar image of the cyclist. Notice that the tilted front wheel of the bicycle is also clearly visible.

4 Future Trend: High Resolution Radar Combined with Artificial Intelligence

The resolution of a radar in a given dimension defines its ability to distinguish between two targets separated by a minimum value in the same dimension. Automotive radars typically measure three different parameters for every target where these parameters are often referred to as dimensions: range, Doppler velocity and angle. For instance, a resolution of 1 m in the range dimension would imply that the radar can distinguish between two targets that are separated by at least a distance of 1 m. If this distance is reduced to any value below the resolution limit, the radar will simply see only a single target. The resolution limit of a radar is determined by its hardware specifications. For example, the amount of time that a radar illuminates a target, determines its Doppler resolution. This means, the longer a moving target is illuminated by the radar signal, the more exact its Doppler velocity components can be estimated.

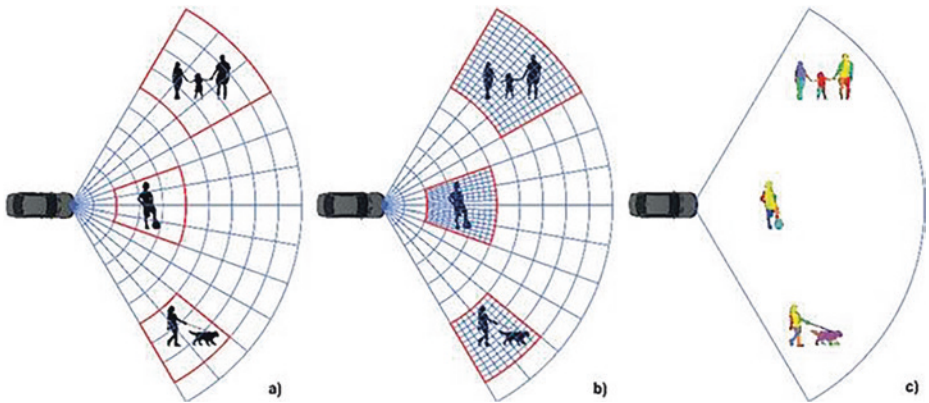


Fig. 9. Two-stage classification algorithm: (a) Stage one identifies regions of interest in regular resolution FoV. (b) The second stage uses high resolution processing on the RoIs for a final object classification as depicted in (c).

A viable alternative to enhancing the hardware in order to improve resolution is to employ high resolution signal processing techniques without modifying the hardware. Like any other compromise, there is a certain give and take. High resolution processing usually requires much more computational power, whereas modifying the hardware for better resolution can keep computational requirements at a lower level, however it increases manufacturing costs. For a transition from radar detection to radar perception, high resolution radars play an integral role. With the availability of highly resolved targets in day-to-day traffic scenarios the task of target classification and identification can be performed with a much higher level of certainty, making the safety of all road users feasible.

As previously mentioned, high resolution algorithms require an increased amount of computational power. A possible approach for solving this problem is to apply these algorithms only within certain regions of the entire traffic scene. This method is often referred to as region of interest (RoI) or frequency selective processing. Figure 8 shows the radar images obtained from an exemplary traffic scenario. The plot in the middle shows data that have been processed using conventional low complexity processing as compared to the plot on the right which has been obtained with the use of high resolution techniques.

To explain this concept, imagine a traffic scenario where pedestrians are walking in a crowded scenario on their own, in groups, or with pets. A typical mid-range automotive radar will be able to see up to 70 m in range and has field of view (FoV) of over 100°. In such a scenario where detection and classification of pedestrians is given top priority, it suffices to apply high resolution for limited ranges (e.g. 10–15 m) and a smaller FoV. The remaining sections of the traffic scenario can be resolved by means of conventional processing with a lower resolution. The incorporation of RoI processing in the classification process is done in a two-step algorithm that uses regular resolution for a classification of RoIs as a first step. In a second step, the highly resolved RoIs are assigned an object class (cf. Fig. 9). The higher resolution enables the classifier to achieve very accurate classifications results; it is even possible to identify different body parts of a pedestrian. This is e.g. crucial for the prediction of gait starting or stopping.

The availability of highly resolved radar images opens the door on further classification and prediction possibilities. In Fig. 8c, the bicycle in the scene is processed using a HR technique [30]. If only low resolution data is available, very few reflections of the bicycle can be measured. However, in the HR processed image, the tilt of the front wheel of the bicycle is clearly exhibited and this information can be used as input for image processing algorithms that try to predict the turning direction of cyclist based on wheel orientation.

5 Conclusion

The paper outlined the challenges for automotive radar to provide the required perception information for autonomous vehicles. It has been outlined that Radar data processing toolchains have to be revolutionized by applying artificial intelligence and advanced model based signal processing in a synergetic manner. From the analysis it became obvious that future automotive radars have to provide extreme resolution in space and velocity.

References

1. Daimler AG: Collision prevention assist. <https://www.mercedes-benz.com/de/mercedes-benz/innovation/mit-abstand-am-besten-assistenzsysteme-von-mercedes-benz/>. Zugegriffen: 26 Februar 2018
2. Daimler AG: Mercedes-Benz-Intelligent-Drive. <https://www.mercedes-benz.com/en/mercedes-benz/innovation/mercedes-benz-intelligent-drive/>. Zugegriffen: 26 Februar 2018
3. Daimler AG: Bertha-Benz Drive. <http://media.daimler.com/marsMediaSite/de/instance/ko/Pionierleistung-Autonome-Langstreckenfahrt-im-Ueberland-und-Stadtverkehr-Mercedes-Benz-S-Klasse-INTELLIGENT-DRIVE-fahrt-autonom-auf-den-Spuren-von-Bertha-Benz.xhtml?oid=9904223>. Zugegriffen: 26 Februar 2018
4. Dickmann, J., Appenrodt, N., Brenk, C.: Bertha fährt autonom. In: Automobil Elektronik, pp. 44–47, March 2014
5. Dickmann, J., Appenrodt, N., Klappstein, J., Bloecher, H.L., Muntzinger, M., Sailer, A., Hahn, M., Brenk, C.: Making Bertha see even ore: radar contribution. *IEEE Access* **3**, 1233–1247 (2015)
6. Dickmann, J., Appenrodt, N., Brenk, C.: Making Bertha See. In: *IEEE Spectrum*, pp. 40–46, August 2014
7. Waymo: Waymo sensor suite. https://cdn-images1.medium.com/max/1600/1*9RQQY5ehT-W5dYAhwdU4PdG.jpeg. Zugegriffen: 26 Februar 2018
8. Uber: Self-Driving-Uber-Vehicles-San-Francisco. <http://www.rideshareconnection.com/self-driving-uber-vehicles-san-francisco/>. Zugegriffen: 26 Februar 2018
9. General Motors: General motors sensor suite. <http://www.repairedrivennews.com/wp-content/uploads/2017/12/general-motors-av-autonomous-sensors57-2-1024x576.jpg>. Zugegriffen: 26 Februar 2018
10. Dickmann, J.: Automotive radar and radar based perception for driver-less cars. In: Ben Gurion University Radar Symposium. Ben-Gurion University of the Negev, Israel (2017)
11. Dickmann, J.: When will we give up driving our cars? A radar developers point of view. In: *ISSCC, San Francisco* (2017)
12. Dickmann, J.: Automotive radar systems at daimler: past-present-future. In: *IEEE Radar Conference, Enabling Technologies for Advances in Radar*, Philadelphia (2016)
13. McCloskey, D.: Radars as a complementary sensor for autonomous driving at Waymo. In: *14th European Radar Conference on EURAD-Workshop: Future Automotive Radar Systems*, Nuremberg (2017)
14. Wyholt, A.: Scalability of sensor systems for active safety and autonomous functions. In: *International VDI Conference – Automotive Sensor Systems*, München (2018)
15. Lombacher, J., Hahn, M., Dickmann, J., Wöhler, C.: Detection of arbitrarily rotated parked cars based on radar sensors. In: *16th International Radar Symposium (IRS)* (2015)
16. Lombacher, J., Hahn, M., Dickmann, J., Wöhler, C.: Object classification in radar using ensemble methods. In: *IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)* (2017)
17. Lombacher, J., Hahn, M., Dickmann, J., Wöhler, C.: Semantic radar grids. In: *IEEE Intelligent Vehicles Symposium* (2017)
18. Schumann, O., Hahn, M., Dickmann, J., Wöhler, C.: Comparison of random forest and long short-term memory network performances in classification tasks using radar. In: *Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, Bonn (2017)
19. Giese, T., Klappstein, J., Dickmann, J., Wöhler, C.: Road course estimation using deep learning on radar data. In: *International Radar Symposium (IRS 2017)*, Prag (2017)

20. Lombacher, J., Hahn, M., Dickmann, J., Wöhler, C.: Potential of radar for static object classification using deep learning methods. In: IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM) (2016)
21. Werber, K., Klappstein, J., Dickmann, J., Waldschmidt, C.: Interesting areas in radar grid-maps for vehicle self-localization. In: International Conference on Microwaves for Intelligent Mobility (ICMIM 2016), San Diego (2016)
22. AdaptIVe Project: AdaptIVe – final project results (2017)
23. Schumann, O., Hahn, M., Dickmann, J., Wöhler, C.: Supervised clustering for radar applications: on the way to radar instance segmentation. In: ICMIM – (2018)
24. Duraisamy, B., et al.: Track level fusion of extended objects from heterogeneous sensors. In: 19th International Conference on Information Fusion (FUSION) (2016)
25. Kellner, D., et al.: Tracking of extended objects with high-resolution doppler radar. IEEE Trans. Intell. Transp. Syst. **17**(5)1341–1353 (2016)
26. Brosseit, P., et al.: The volcanormal density for radar-based extended target tracking. In: IEEE 20th International Conference on Intelligent Transportation Systems (ITSC) (2017)
27. Duraisamy, B., et al.: Combi-Tor: track-to-track association framework for automotive sensor fusion. In: IEEE 18th International Conference on Intelligent Transportation Systems, Las Palmas (2015)
28. Duraisamy, B., et al.: Track level fusion algorithms for automotive safety applications. In: International Conference on Signal Processing, Image Processing & Pattern Recognition (2013)
29. Sarholz, F., et al.: Evaluation of different quality functions for road course estimation using imaging radar. In: Intelligent Vehicles Symposium (IV) (2011)
30. Dehkordi, S., Appenrodt, N., Dickmann, J., Waldschmidt, C.: Region of interest based adaptive high resolution parameter estimation with applications in automotive radar. In: IRS, Bonn (2018)
31. Duraisamy, B., et al.: Influence of the sensor local track covariance on the track-to-track sensor fusion. In: IEEE 18th International Conference on Intelligent Transportation Systems (2015)
32. Duraisamy, B., et al.: Object management strategy for a unified high level automotive sensor fusion framework. In: IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI), Baden-Baden (2016)
33. Mayer, W.: Abbildender Radarsensor mit sendeseitig geschalteter Gruppenantenne, Dissertation. Institut fuer Mikrowellentechnik, University of Ulm, Ulm (2008)
34. Daimler AG: Remote parking pilot. <http://media.daimler.com/marsMediaSite/de/instance/ko/Remote-Park-Pilot-Ferngesteuertes-Parken-per-Smartphone-App.xhtml?oid=9361355>. Zugegriffen: 26 Februar 2018
35. European Project MOSARIM: MOre Safety for All by radar Interference Mitigation, Proj. Ref. No: 248231, FP7-ICT (2014)
36. Muntzinger, M., Aeberhard, M., Zuther, S., Schmid, M., Dickmann, J., Dietmayer, K.: Reliable automotive pre-crash system with out-of-sequence measurement processing. In: IEEE Intelligent Vehicles Symposium, pp. 1022–1027 (2010)
37. Andres, M., Feil, P., Menzel, W.: 3D-scattering center detection of automotive targets using 77 GHz UWB radar sensors. In: EuCAP, Prag (2012)
38. Hosseini, A., Diewald, F., Klappstein, J., Dickmann, J., Neumann, H.: Modification of the landweber method based on the conjugate gradient method to restore automotive radar images. In: International Conference on Systems, Signals and Image Processing (IWSSIP), Wien (2012)
39. Kellner, D., Barjenbruch, M., Klappstein, J., Dickmann, J., Dietmayer, K.: Instantaneous full-motion estimation of arbitrary objects using dual doppler radar. In: Intelligent Vehicle Symposium (IV 2014), Dearborn (2014)

40. Sarholz, F., Mehnert, J., Klappstein, J., Dickmann, J., Radig, B.: Evaluation of different approaches for road course estimation using imaging radar. In: *Intelligent Robots and Systems* (2011)
41. Dickmann, J., Klappstein, J., Hahn, M., Muntzinger, M., Appenrodt, N., Brenk, C., Sailer, A.: Present research activities and future requirements on automotive radar from a car manufacturer's point of view. In: *IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)*, Heidelberg (2015)
42. Dickmann, J., Meinel, H.: Automotive radar: from its origins to future directions. *MWJournal* **56**(9), 24–40 (2013)